Derivation appendix: 02 — S_3 fermion sector

the Yukawa dictionary, and the CKM obstruction

1. The draft's S_3 doublet basis is feynlag's reflected at $\pi/6$; the Yukawa coupling dictionary is the identity

derived here — §1

Independent checks

- O conjugates feynlag's generators into the draft's, $\det O = -1$
- each triple-doublet form is invariant only in its own basis
- the draft's triple structure maps to feynlag's with coefficient $+1$

Would be falsified by: a coefficient other than $+1$ under the map, or a triple form invariant in both bases

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2. The $\mu \leftrightarrow Y$ dictionary, and the draft's factor $\sqrt{2}$

derived here — §4.1

Claim

$$\mu_k^\ell = \frac{Y_k^\ell v}{\sqrt{2}} \text{ (plain Lagrangian)}, \qquad \mu_k^\ell = \frac{Y_k^\ell v}{2} \ \ (k \neq 3) \text{ with the draft's prefactors}$$

Result

$$\begin{bmatrix} \mu_1 & \frac{Y_1^\ell v_S}{2} & \frac{\sqrt{2}Y_1^\ell v_S}{2} \\ \mu_2 & \frac{Y_2^\ell v_1}{2} & \frac{\sqrt{2}Y_2^\ell v_1}{2} \\ \mu_3 & \frac{\sqrt{2}Y_3^\ell v_S}{2} & Y_3^\ell v_S \\ \mu_4 & \frac{Y_4^\ell v_1}{2} & \frac{\sqrt{2}Y_4^\ell v_1}{2} \\ \mu_5 & \frac{Y_5^\ell v_1}{2} & \frac{\sqrt{2}Y_5^\ell v_1}{2} \end{bmatrix}$$

Independent checks

- read off the mass matrix rather than typed in
- the same factor in all five, so a convention not a per-term slip
- restoring $\langle H^0 \rangle = v$ reproduces the printed values exactly

Would be falsified by: a ratio differing between the five μ's, which would mean a genuine per-term error rather than a normalisation

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3. $(\sigma_1 - \sigma_2)^2 = (\rho - \mu_3)^2 + 4(\mu_4 + \mu_5)^2$, not $+16\mu_4\mu_5$

derived here — §5.1

$$(\sigma_1 - \sigma_2)^2 = 4\,(\mu_4 + \mu_5)^2 + (\mu_1 + 2\mu_2 - \mu_3)^2$$

Independent checks

- built from $\mathrm{tr}(A^\mathsf{T}A)$ and $\det A$, neither closed form typed in
- symmetric under $\mu_4 \leftrightarrow \mu_5$, as singular values require
- at $\mu_5 = 0$ the two forms differ by $4\mu_4^2$

Would be falsified by: the printed form reproducing $\mathrm{tr}(A^\mathsf{T}A) - 2\det A$ for independent μ_4, μ_5

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4. The first generation decouples iff $r = v_1/v_2 = \sqrt{3}$ — the $\mathbb{Z}_2$-preserving vacuum — hence $V_{us} = 0$ with exact S_3

derived here — §10

Claim

$$\sqrt{v_1^2 + v_2^2} = 2\,|v_2| \iff |v_1| = \sqrt{3}\,|v_2| \text{ (draft basis)}$$

Result

$$r\left(2 - \sqrt{r^2 + 1}\right) = 0$$

Independent checks

- eigenvector taken from eigenvects(), not written down
- decoupling recast as orthogonality to the vacuum direction $(r, 1)$
- the $(1,3)$ entry is non-zero at four other values of r
- O_{12} diagonalises the 1–2 block for arbitrary r, so the same rotation serves every fermion sector

Would be falsified by: a second positive root, or a non-zero $(1,3)$ entry at $r = \sqrt{3}$

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5. Soft S_3 breaking reproduces the six quark masses and $|V_{us}|$, $|V_{cb}|$, $|V_{ub}|$ simultaneously (an existence result, not a prediction)

derived here — §11.2

Independent checks

- all nine targets reproduced at a fit cost below 10^{-20}
- at least four distinct exact solutions found (the count depends on the numpy/scipy version); the one kept is selected by a stated rule, |r| closest to sqrt3
- $|V_{us}|$ vanishes exactly at $r = \sqrt{3}$ and grows linearly away from it

Would be falsified by: no exact solution for $r \neq \sqrt{3}$, or one reproducing $|V_{us}|$ at $r = \sqrt{3}$

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6. The S_3 lepton assignment and its five Yukawa structures

imported — §2–3 (from [LFVHD])

Independent checks

- suggest_yukawa independently returns exactly 5
- all five pass gauge and S_3 invariance
- a deliberately mis-contracted Y_2 fails the S_3 check

Would be falsified by: suggest_yukawa returning a different count

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